# Section 13: Cisco Troubleshooting Methodology

Layer: All | Reference: Odom Vol2 Ch31: Troubleshooting Tools | *Original study notes — CCNA 200-301*

A consistent troubleshooting methodology turns a stressful outage into a repeatable process. CCNA exam simlets reward systematic layer-by-layer elimination over guesswork.

## Key Concepts

- Bottom-up approach: start at Layer 1 (cabling/link lights), work upward through L2/L3/L4 - best when the cause is unknown and could be anywhere
- Top-down approach: start at the application/user-reported symptom and work downward - best when you suspect a specific high-layer service
- Divide-and-conquer: jump to a middle layer (commonly L3) first, then go up or down based on whether that layer works - efficient when you have a hunch
- Follow-the-path: trace the actual physical/logical path the traffic takes hop by hop, checking each device along the way
- Compare-configuration: diff a known-working config against the current one to spot an unintended recent change
- Always verify Layer 1 first in any methodology when physical connectivity is in question - 'show interfaces' and link lights catch the most common simple failures
- Document baseline 'normal' behavior before issues occur - without a baseline, 'is this normal?' is unanswerable during an incident
- Change control discipline: most outages trace back to a recent change; always ask 'what changed?' before assuming hardware failure
- Escalation and ticket documentation matter for the exam's process-based questions, not just the technical fix itself

## Command Reference

| Command | Purpose |
|---|---|
| `show interfaces` | Check Layer 1/2 status, errors, drops, and utilization on an interface |
| `show ip interface brief` | Quick up/down and IP-assignment status across all interfaces |
| `show cdp neighbors` | Verify physical topology and confirm a cable is connected to the expected device |
| `ping / traceroute` | Test reachability and trace the path layer by layer (L3) |
| `show logging` | Review recent system messages, including interface flaps and security violations |

| Command | Purpose |
|---|---|
| `show tech-support` | Generate a comprehensive snapshot for escalation or deep analysis (very long output) |

## Configuration Walkthrough

**Typical bottom-up troubleshooting sequence for 'PC can't reach the server'**

```
! 1. Physical: check link light, show interfaces (look for up/up, no errors)
! 2. Data Link: show interfaces switchport, show mac address-table
! 3. Network: show ip interface brief, ping default gateway
! 4. Transport+: ping/traceroute to the server, check ACLs along the path
```

## Worked Scenario

**Q: A user reports 'the network is down' on their PC. What's a defensible first diagnostic step under a bottom-up methodology?**

Check Layer 1 first: verify the physical link light status on both the PC's NIC and the switch port (or run 'show interfaces' on the switch to confirm line protocol is up). A huge share of real-world 'network down' reports trace back to a disconnected or damaged cable, a disabled switch port, or a duplex/speed mismatch - and ruling this out takes seconds compared to chasing routing or DNS issues first.

## Common Mistakes / Gotchas

- Jumping straight to advanced Layer 3/4 troubleshooting without confirming Layer 1/2 basics first
- Not asking 'what changed recently?' before assuming a hardware fault
- Skipping documentation of the baseline/normal state, making it harder to identify what's actually abnormal

## Exam Tips

★ *Know the names and use-cases for each methodology: bottom-up, top-down, divide-and-conquer, follow-the-path, compare-config*

★ *Simlet troubleshooting tasks often reward a structured layer-by-layer approach over random command guessing*